

EDUCATION

- **Chandigarh University** Mohali, Punjab
B.Tech in Computer Science & Business Systems; CGPA: 8.27/10 Aug 2023 – Jun 2027
 - **Relevant Coursework:** Data Structures, Computer Architecture, Operations Research, Software Engineering, Statistics, Linear Algebra

EXPERIENCE

- **IIT Roorkee** Uttarakhand, India
SPARK Research Intern May 2025 – June 2025
 - **ML Pipeline:** Built and deployed an end-to-end ML pipeline, automating data ingestion, training, and evaluation stages to cut manual processing overhead.
 - **Performance:** Reduced prediction error to 11.5% through systematic experimentation, cross-validation, and pipeline tuning.
 - **Documentation:** Documented infrastructure and methodology in technical reports for reproducibility across research stakeholders.
 - **Tech Stack:** Python, Pandas, NumPy, Scikit-learn, Power BI
- **DeepLearning.AI** Remote
QA Tester — AI/ML Specializations June 2023 – April 2025
 - **Quality Assurance:** Tested datasets, coding exercises, and evaluation pipelines across cloud-hosted AI/ML specializations, validating environment configuration and execution reliability.
 - **Collaboration:** Identified bugs and pipeline failures, collaborating asynchronously with distributed global teams to improve platform stability.
 - **Tech Stack:** Python, Git, GitHub
- **Divine Himalayan Foundation** India
Operations Intern May 2024 – July 2024
 - **Payment Integration:** Integrated Razorpay payment infrastructure for international donation processing, including API configuration, webhook testing, and operational workflow support.
 - **Data Analysis:** Performed comprehensive donor data analysis leading to a 200% increase in fundraising revenue.
 - **Tech Stack:** Python, Excel, Razorpay

PROJECTS AND PUBLICATIONS

- **ML-Optimized Stochastic VRP for Urban Delivery (ACROSET 2026)** 2026
Optimization & Benchmarking
 - **Framework:** Built an end-to-end Chandigarh SVRP pipeline integrating OR-Tools, OSMnx, and metaheuristic solvers (ABC, ACO, ALNS) with demand and traffic forecasting models (ARIMA, LSTM, SPO+).
 - **Evaluation:** Designed a Monte Carlo scenario simulator to stress-test solver robustness; OR-Tools achieved lowest mean delivery cost (4.09% improvement over worst solver) and ALNS reduced mean lateness by 12.34%.
 - **Tech Stack:** Python, OR-Tools, OSMnx, ARIMA, LSTM, GitHub
- **Delivery Route Optimization (ICCTDC 2025)** 2025
Optimization Algorithms
 - **Optimization:** Developed a Vehicle Routing Problem solver using the Artificial Bee Colony Algorithm, achieving a 33% reduction in delivery time.
 - **Impact:** Research accepted and published at ICCTDC 2025.
 - **Tech Stack:** Python, Pandas, Optimization Algorithms
- **Swasth AI — Mental Health Support Platform** 2024
AI/ML Application
 - **Development:** Built and deployed an AI-assisted platform integrating conversational AI and ML-based stress detection.
 - **Leadership:** Led team coordination, sprint planning, and infrastructure setup for project demonstrations.
 - **Achievement:** Won 1st Place in a university hackathon among 1,000+ participants.
 - **Tech Stack:** Python, ML, Conversational AI

SKILLS & AFFILIATIONS

- **Cloud:** AWS Certified Cloud Practitioner; EC2, S3, Lambda, IAM, CloudWatch
- **Infrastructure & Automation:** Git, GitHub Actions, Linux Administration, Bash Scripting, CI/CD Fundamentals
- **Programming Languages:** Python, C++, SQL, Bash
- **Data & Backend:** PostgreSQL, ETL Pipelines, REST APIs, Pandas, NumPy, Scikit-learn, NoSQL
- **Visualization :** Grafana, Power BI, Streamlit
- **Engineering Practices:** Version Control, Monitoring & Logging, Performance Benchmarking, A/B Testing
- **Leadership & Memberships:** Tech Lead, Chandigarh University Astronomy Club; IEEE Delhi Section Member